

ex 801...
⇒

$$\int_0^1 \frac{16n}{8n^2+2} dn$$

as per rule no 2L

$$\left\{ \frac{f(n)'}{f(n)} = \ln n + c \right.$$

sol

$$\int_0^1 \frac{16n}{8n^2+2}$$

$$\int_0^1 (\ln(8n^2+2)) + c$$

Now Apply Limit:- [0 to 1]

$$\left[\ln(8n^2+2) \right]_0^1 = \ln(8(1)^2+2) - \ln(8(0)^2+2)$$

$$= \ln(10) - \ln(2)$$

$$= \ln\left(\frac{10}{2}\right) = \ln(5)$$

$$\textcircled{2} \int \frac{x^2}{x^2+1} dx$$

so

~~$$\int x^2 dx \times \int \frac{1}{x^2+1} dx$$~~

~~$$\frac{x^3}{3} \times \frac{1}{2}$$~~

step 1,
add 1 and -1

$$\int \frac{x^2+1-1}{x^2+1} dx$$

in numerator

$$\int \frac{(x^2+1)}{x^2+1} dx - \int \frac{1}{x^2+1} dx$$

$$= \int dx - \int \frac{1}{x^2+1} dx$$

Now we will use the substitution method to find the integral of right hand side -

~~$$\int dx$$~~

$$\begin{aligned} &\text{for } x^2+1 \\ &\text{let } x = \tan \theta \\ &dx = \sec^2 \theta d\theta \end{aligned} \quad \left| \quad \frac{d}{dx} = \tan \theta \right.$$

$$= \frac{1}{\tan^2 \theta + 1} \cdot \sec^2 \theta d\theta$$

$$\tan^2 \theta + 1$$

Note: $\boxed{\tan^2 \theta + 1 = \sec^2 \theta}$ that's why we can cancel it out.

$$= \frac{1}{\cancel{\tan^2 \theta + 1}} \cdot \cancel{\sec^2 \theta} d\theta$$

$$= \int d\phi$$

So

$$= \int dn - \int d\phi$$

$$= n - \phi$$

Now we know that

$$n = \tan \phi$$

so

$$n = \arctan n + c$$